

Data Preparation Mock Exam Questions

Question 1: What is wrong with the following code? How would you fix it?

```
df <- data.frame(A = c(1, 2, NA, 4), B = c(5, 6, 7, 8))  
df <- filter(df, !is.na(A | B))
```

Question 2: What is wrong with this regular expression for extracting email addresses? Fix the mistake.

```
emails <- str_extract_all(text, "[a-zA-Z0-9_]+@[a-zA-Z_]+\\.[a-zA-Z]+")
```

Question 3: Fill in the missing function to replace all missing values in column Price with the mean of that column:

```
df$Price[is.na(df$Price)] <- _____(df$Price, na.rm = TRUE)
```

Question 4: Match the regular expression pattern to its purpose.

Regex Pattern	Purpose
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\\d+	(a) Extract email addresses
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\\w+@\\w+\\.\\w+	(b) Extract numbers
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[A-Z]{3}	(c) Find three-letter uppercase codes
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Question 5: Match each R function to the type of data it is used to load.

Function	Data Type
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A. read.csv()	1. JSON File
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B. fromJSON()	2. CSV File
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C. readRDS()	3. Excel File
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D. read_excel()	4. R Serialized Object
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Question 6: Match each function to its correct purpose.

Function	Purpose
A. pivot_longer()	1. Convert wide format to long format
B. pivot_wider()	2. Convert long format to wide format
C. separate()	3. Split a column into multiple columns
D. unite()	4. Merge multiple columns into one

Question 7: Match the method with the correct way it identifies outliers.

Method	Description
A. boxplot.stats(df\$column)\$out	1. Identifies outliers using the interquartile range (IQR) method
B. abs(scale(df\$column)) > 3	2. Identifies outliers based on standard deviation (z-score method)
C. df %>% filter(df\$column > mean(df\$column, na.rm = TRUE) + 3 * sd(df\$column, na.rm = TRUE))	3. Filters extreme values beyond 3 standard deviations from the mean
D. quantile(df\$column, probs = c(0.05, 0.95), na.rm = TRUE)	4. Finds the 5th and 95th percentile to detect potential outliers

Question 8: You receive a dataset df with duplicate rows. How would you remove duplicates while keeping only the first occurrence?

Question 9: A dataset contains timestamp data in different time zones. Explain how you would standardize all timestamps to UTC in R.

Question 10: What will be the result of the following filtering operation?

```
df <- data.frame(A = c(10, 15, 20, 25), B = c(5, 7, 10, 12))  
df_filtered <- filter(df, A > 15 & B < 10)  
print(df_filtered)
```

Question 11: Predict the output of this R command:

```
str_extract_all("My number is 123-456-7890", "\\d{3,}")
```

Question 12: What does this command return?

```
mean(c(10, 20, 30, NA), trim = 0.2, na.rm = TRUE)
```

Question 13: Write a regex pattern to find dates in either YYYY-MM-DD or DD/MM/YYYY format.

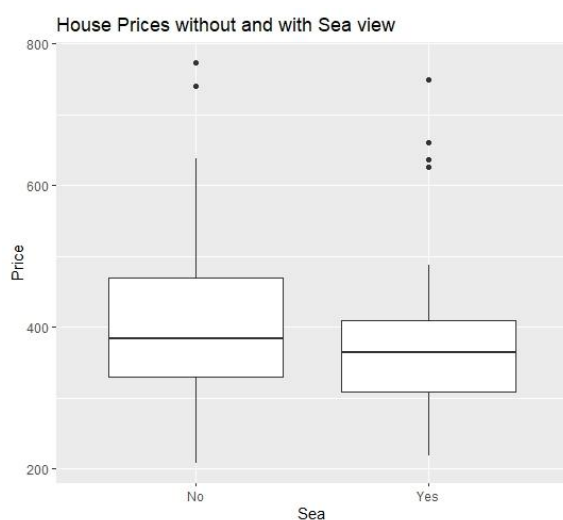
Question 14: Write a regex pattern to extract website URLs that start with http:// or https://.

Question 15: Write a regex pattern to extract all words starting with a capital letter from a text string.

Question 16: A dataset contains missing values in multiple columns. Describe two approaches to handle them and explain when each approach is appropriate.

Question 17: You have a dataset with Sales values that contain extreme outliers. Describe a robust way to handle them without removing data.

Question 18: Our dataset is about house prices in a small city near the sea. The data is coming from a real estate agency, and includes the land (in hectares), the size of the house (square meter), the age of the house (year), the number of rooms, bathrooms, garages and whether the house has a sea view.



How many outliers are among the houses with sea view?